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Fostering Shared Understanding for Effective Learning through Conversational AI

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ABSTRACT

Although conversational agents are successfully applied in teaching, it is largely unclear which communication principles should be employed to optimise learning. We examine the influence The control version provided information without emphasising grounding, whereas the pedagogical agents; common ground version emphasised grounding, for example, by encouraging students to chatbots; common ground; monitor and repair common ground. After the learning unit, students evaluated their learning higher education experience and the pedagogical conversational agent, after which they were tested on the studied material. Students in the common ground (vs. the control) condition performed better in a post-study knowledge test and engaged longer with the pedagogical conversational agent. Thus, the common ground emphasis facilitated learning with a conversational agent, indicating that grounding principles should be incorporated when designing conversational agents.

1. Introduction

Compared to one-size-fits-all teaching, customisation facilitates learning (e.g. Bernacki, Greene, and Lobczowski 2021; Tomlinson et al. 2003). However, in higher education, the numbers of students per instructor is usually high (OECD 2022), preventing adaptations to individual learners. This problem can be solved by employing pedagogical conversational agents (PCAs), conversational agents that act as digital tutors. PCAs can adapt to individual learners and thus provide tailored teaching, making one-on-one teaching possible even for large numbers of students (Mousavinasab et al. 2021; Tavangarian et al. 2004). Previous research has already demonstrated that PCAs can foster learning (Kulik and Fletcher 2016; Wu and Yu 2024). However, it is largely unclear how they should communicate in order to optimise learning outcomes. Accordingly, improving communication has been named the most promising research direction for PCAs (Sikström et al. 2022).

The present research applies a key principle from the psychology of communication to pedagogical conversational agents – establishing *common ground*. Common ground describes shared knowledge to which conversation partners can refer during communication (Clark 1996). The establishment of common ground is crucial for efficient information transmission (Vygotsky 1986), especially in teaching (Alibali et al. 2019). In the present field experiment, students interacted with a PCA that emphasised grounding or with a PCA that provided identical explanations without emphasising grounding. We examined whether this emphasis on common ground enhances learning in an applied higher education setting. Specifically, we asked the following research question: Does the incorporation of grounding strategies influence students' learning experiences and learning success with a pedagogical conversational agent?

2. Theoretical background

2.1. Pedagogical conversational agents

Conversational Agents – such as chatbots, virtual agents, and artificial conversational entities – are software applications that facilitate dialogue between humans and computers (for reviews, see e.g. Adamopoulou and Moussiades 2020; Kiuchi, Otsu, and Hayashi 2024). Conversational Agents use natural language processing to recognise, interpret, and respond to user input, thereby improving human – machine interactions (Gnewuch, Morana, and Maedche 2017; Serban et al. 2017). Because of their high

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flexibility and user- friendliness, conversational agents are employed for various purposes – for example, to provide technical support (e.g. Subramaniam et al. 2018), to conduct interviews (e.g. Sidaoui, Jaakkola, and Burton 2020), and to act as service-assistants (Maedche et al. 2016) – and in multiple domains – for example, commerce (Graham et al. 2025; Hari, Iyer, and Sampat 2022; Voorveld et al. 2024), health care (Brown et al. 2024; Fitzpatrick, Darcy, and Vierhile 2017; Roca et al. 2020), and entertainment (Wilkinson et al. 2021).

In conversational agents, communication styles that include humanising and social elements, instead of styles focused purely on information exchange, have been found to be beneficial (Araujo 2018; Jiang, Yang, and Zheng 2023; Thomas et al. 2021). For example, adding anthropomorphic elements, such as small-talk and first-person pronouns, to the communication of a conversational agent improved user evaluations (Qu, Zhou, and Chen 2022), satisfaction (Konya-Baumbach, Biller, and von Janda 2023), trust (Munnukka, Talvitie-Lamberg, and Maity 2022), and compliance (Adam, Wessel, and Benlian 2021). Similarly, conversational agents that employed relationship-building and empathic elements improved user evaluation and engagement with the conversational agent (Bickmore and Picard 2005; Brunswicker et al. 2025; Janson 2023; Kim and Im 2023), so that generally, a socially-oriented instead of task-oriented conversational style has been found to be beneficial (Pecune, Callebert, and Marsella 2022). Thus, how conversational agents communicate impacts user evaluations and behaviour.

Humans frequently treat conversational agents as, or similar to, social actors (Feine et al. 2019; Gambino, Fox, and Ratan 2020). According to Social Response Theory, humans apply social norms to their interactions with conversational agents, anticipating appropriate understanding and responses (Nass and Moon 2000; Reeves and Nass 1996). Humans have been found to attribute, for example, personality (Ahmad, Siemon, and Robra- Bissantz 2021) and mental abilities (Weisman, Dweck, and Markman 2017) to conversational agents. Moreover, similar to conversations with humans, people adopt terms used by computers (Branigan et al. 2010; Shen and Wang 2023) and adhere to politeness norms (Nass, Moon, and Carney 1999). Accordingly, although differences between human – computer and human – human communication have also been found (e.g. Hill, Ford, and Farreras 2015; Shen and Wang 2023) insights from the psychology of communication could be used to improve human – computer interactions (Torrey, Fussell, and Kiesler 2013).

A major and growing field for conversational agents is education (for reviews, see Kuhail et al. 2023; Pérez, Daradoumis, and Puig 2020; Winkler and Söllner 2018). Besides efficiency of education, the main aim of PCAs is improving students' motivation and study skills and ultimately academic success (Wollny et al. 2021). To achieve these aims, PCAs can provide individualised assistance and thereby make education more interactive and adaptive (Ondáš, Pleva, and Hládek 2019; Sibley et al. 2024). They can make learning more student-guided by enabling learners to choose when and how to respond and then adapt to individual student behaviour (Studente and Ellis 2020). Moreover, PCAs can be employed to stimulate self-regulation, thereby reducing procrastination (Pereira and Díaz 2022). In general, one major advantage of PCAs is that they can adapt learning content and instructional mode to individual preferences and styles (Filho et al. 2021; Gubareva and Lopes 2020).

Behavioural and communication strategies employed in pedagogical conversational agents can foster or hinder learning success. PCAs with enthusiastic and encouraging (vs. neutral) behaviour have been found to increase student motivation and engagement, leading to better learning outcomes (Baylor and Kim 2005). Moreover, and similar to conversational agents in other domains (Bickmore and Picard 2005; Konya- Baumbach, Biller, and von Janda 2023), a personalised communication style (Lin et al. 2020), polite responses (Wang et al. 2008), as well as the inclusion of affect (Guo and Goh 2015) and empathic responses (Kim, Baylor, and Shen 2007) can be beneficial in learning situations.

In computer-assisted teaching, several teaching strategies have been found to improve learning. PCAs that provide immediate and specific feedback after participants' answering test questions improved learning by addressing specific misconceptions (Arroyo et al. 2014) and by increasing behavioural engagement (Schiller et al. 2024). Such feedback was most effective when it pertained to individual steps (VanLehn 2011), provided explanations for the correction of errors (Narciss and Huth 2006), and when it was detailed (Van der Kleij, Feskens, and Eggen 2015). Another efficacious teaching strategy is scaffolding, a gradual increase in complexity (Duffy and Azevedo 2015; for a meta-analysis see, Hooshyar et al. 2024). Several studies found that PCAs that included (vs. did not include) scaffolding improved learning success (e.g. Lin and Hung 2025; Winkler et al. 2020). In general, teaching

strategies that facilitate learning with human teachers also have been found to facilitate learning with PCAs (Zhang et al. 2024). Thus, conversation principles that help human communication could also improve human – computer interaction, and more specifically learning with PCAs. The present research tests whether employing the human communication principle of common ground establishment facilitates learning with PCAs.

2.2. Common ground

Common ground refers to knowledge and beliefs that human conversation partners assume they share and therefore use in communications (Clark and Brennan 1991; Clark and Marshall 1981). For example, in a matching task, conversation partners typically establish terms or phrases to refer to specific objects; and the usage of these referential terms improves task performance (Clark and Wilkes-Gibbs 1986; Fussell and Krauss 1989a). Common ground consists in information that is co-present (e.g. objects in a common visual field), information mentioned during the present or past interactions, and information that is assumed to be shared within a shared community or culture (Gann and Barr 2014; Lau, Chiu, and Hong 2001). Accordingly, messages created for a friend compared to a stranger are shorter and lead to superior performance (Fussell and Krauss 1989b; Nault et al. 2023), indicating that communication partners use common ground that was established in previous interactions and that doing so improves communication. Similarly, information assumed to belong to a cultural common ground is also referred to with shorter phrases (Fussell and Krauss 1992; Tobar-Henríquez, Rabagliati, and Branigan 2021). Thus, human conversation partners typically achieve and maintain common ground, which facilitates successful communication.

Several conversational strategies facilitate the establishment and maintenance of common ground (Dideriksen et al. 2023). Senders perform audience design, that is, they create messages that they assume will match the receiver's requirements (e.g. Clark and Murphy 1982; Horton and Gerrig 2005). For example, senders gave longer explanations when referring to less (vs. more) well-known objects (Fussell and Krauss 1992), and teachers gave more detailed explanations to younger compared to older students (Turner and Knutsen 2021). Similarly, medical experts used less technical language when answering questions that also used everyday (vs. technical) language (Bromme, Jucks, and Wagner 2005), showing that message features, such as length and technical terms, are adapted for receivers.

Receivers' key task in conversations is determining when new information is understood well enough for a given purpose (Clark and Schaefer 1989; Micklos and Woensdregt 2023). Thus, receivers decide whether they accept presented information as sufficiently understood or they require additions or clarifications. To signal understanding, receivers can provide backchannels, that is, acknowledgments (such as 'ok' or non-verbal signals), paraphrases, or verbatim repetitions (Brunner 1979; Yngve 1970). Not interrupting the sender but showing continued attention can also signal sufficient understanding (Clark and Brennan 1991). To signal insufficient understanding, receivers can initiate or request repairs (Dingemanse et al. 2015), for example, by asking clarification questions (Schegloff, Jefferson, and Sacks 1977). Compared to affiliative conversations, task-oriented conversations have been found to elicit more repairs, indicating that receivers adapt their communication to current conversational requirements (Dideriksen et al. 2023).

Teaching requires establishing common ground between learner and teacher concerning the learning materials (Isaacs and Clark 1987). Teaching in tertiary education typically involves multiple students per instructor, so that grounding has to occur between multiple people. When crafting a message for two receivers, senders typically adapt to the common ground shared with the less knowledgeable receiver (Yoon and Brown-Schmidt 2014), whereas for larger receiver groups, messages are typically adapted to a medium amount of common ground (Yoon and Brown-Schmidt 2019). Moreover, task performance was found to be inferior when receiving explanations made for someone else compared to oneself (Schober and Clark 1989). Thus, unless it can be adapted to each student individually, messages will presuppose too much or too little common ground for some students. By attempting the establishment of common ground with multiple students at once, traditional teaching leads to inefficient communication. To prevent this problem, conversational agents can be employed in teaching to provide individualised learning conversations.

2.3. The current study

The present research targets the question how pedagogical conversational agents should communicate in higher education. Based on the rich literature

indicating that common ground is essential in human communication, we tested whether common ground principles also benefit studying with a PCA. Although grounding has been postulated to be a key conversational principle not only for human – human but also for human – computer interaction (Brennan 1998; Collins et al. 2024; Rothwell, Shalin, and Romigh 2021), empirical research is rare (Wray, Kirk, and Folsom-Kovarik 2022; for examples, see Müller, Paul, and Li 2021; Peña et al. 2023). Here, we examined whether facilitating grounding in interactions between PCAs and students improves students' test performance.

For the sake of experimental rigour, the present manipulation consists only of encouraging receiver-sided grounding strategies while holding sender-sided grounding strategies (especially audience design) constant. That is, all participants received identical information and were offered identical additional explanations by the PCA. The common ground version of the PCA differed from the control version in three aspects. First, the common ground version provided more information about itself, thereby facilitating a more accurate mental model. Second, the common ground version repeatedly invited participants to think about and judge whether or not there was common ground with the PCA concerning central aspects of the learning materials, and then to either acknowledge or repair common ground. Third, the common ground version invited participants to think about and voice open questions and thereby describe gaps in common ground. This design choice allowed us to isolate the effect of prompting students to reflect on grounding, making causal inferences about this mechanism more robust compared to alternatives that combine multiple grounding manipulations.

The two versions of the PCA were evaluated in a field-setting, specifically in a university course. Students were using the conversational agent in class to learn course materials for an introductory psychology course and were tested for learning success (Smutny and Schreiberova 2020). We tested whether the grounding intervention improved learning. We hypothesised (1) that prompting participants to monitor grounding, provide acknowledgements, and initiate repairs would improve common ground, so that participants who received (vs. did not receive) grounding questions would have a larger experienced common ground with the pedagogical conversational agent. Crucially, we also hypothesised (2) that participants who studied with the common ground version would show better post-study test performance than participants who studied with the control version of the pedagogical conversational agent.

3. Method

All data, materials and analyses are available at https://osf.io/xpt8y/. The study was not pre-registered. Ethics approval was granted by the local ethics committee (No. zEK-22).

3.1. Participants

Psychology students who were enrolled in an introductory psychology course were invited to participate in class. The study was part of a course unit about philosophy of science. 203 students in three successive cohorts started the study and 145 completed all questions.[1] We excluded participants who interacted for less than three minutes with the pedagogical conversational agent, leaving a final sample of 129 students[2] (M_{age} = 21 years, SD_{age} = 4 years; 94 female, 32 male, 3 preferred not answer the gender question). Sample size was determined by the number of students in three successive cohorts completing the study and meeting inclusion criteria. A sensitivity analysis indicates that the present sample had 80% power (with α = .05) to detect a one- tailed effect size of d = 0.44 in the between-group comparison.

We checked whether the two groups differed in baseline performance or other measures that are independent of the present manipulation. Concerning baseline performance, participants in the common ground condition (M = 9.64, SD = 1.95) did not differ from participants in the control condition (M = 9.70, SD = 1.54), $t(123) = 0.20$, $p = .84$, $d = -0.04$, 95% CI = [−0.38, 0.31]. There is also no significant difference for personal innovativeness $t(126) = 0.16$, $p = .88$, $d = -0.03$, 95% CI = [−0.37, 0.32], nor for previous chatbot usage, $\chi^2(1) = 1.62$, $p = .20$.

3.2. Pedagogical conversational agents

The pedagogical conversational agent had the role of a tutor. Interaction was purely text-based, see example conversations in Figures A1–A3. The PCA adopted a friendly but information-oriented conversational style without including affect or empathy.

For both versions of the PCA[3], the learning activity consisted in three loops, each starting with a short textbook-type text (287–312 words long), followed by the option to receive additional information. After

receiving any requested additional information, participants were informed that they could revisit the initial text; otherwise or afterwards, a quiz question about the newly studied content ensued. Participants answered the question and received feedback with an additional explanation (see Wisniewski, Zierer, and Hattie 2020); then the next loop ensued.

The two versions of the PCA differed in three aspects (for examples of each, see Appendix). First, the common ground version included a longer introduction (see Figure A1). The control PCA introduction consisted only of a greeting, a one-sentence summary of the upcoming topic and a one-sentence instruction of how to interact with the PCA. The common ground PCA additionally contained an explanation, one short paragraph, of its aims and interaction capacities, to facilitate participants' forming an accurate mental model, which is important for grounding (Kiesler 2005; Tolzin and Janson 2025).

Second, in each loop, the wording when asking participants to decide whether or not to receive additional information differed (see Figure A2). In the control version, the participants were asked whether they wanted additional information and examples on the topic and were informed that, otherwise, they would be immediately preceding to a quiz question. In the common ground version, this question was split up into three questions, each asking whether participants had sufficient common ground with the PCA on one key concept of the preceding text (e.g. 'Do you know what a causal mechanism is?') or alternatively wanted additional information. Participants were not asked to provide any explanations but simply to think about whether there was sufficient common ground and accordingly provide a backchannel or initiate a repair (Corti and Gillespie 2016; Dideriksen et al. 2023). When participants indicated sufficient common ground, they received an acknowledgement, followed by the next question. When participants initiated a repair, they were given an additional example and explanation. This was followed by an acknowledgement, transitioning to the next question. The additional information participants received upon request was identical for both versions of the PCA.

Third, the end of each loop (after receiving feedback on the quiz question) differed between the two versions of the PCA (see Figure A3). The common ground PCA but not the control PCA invited participants to consider whether anything remained unclear about the topic and offered the possibility to pose questions (initiating a repair; Dideriksen et al. 2023). In the common ground condition, 8 questions were supplied by participants (e.g. 'Are ad hoc assumptions simply exceptions?'). These questions were not answered by the PCA. Instead, participants received an acknowledgement that the question would be passed on to the course instructor.

3.3. Procedure

Before interacting with the chatbot, participants provided informed consent and answered five multiple choice baseline knowledge questions, with a total of 16 dichotomous items. All parts of the study were self-paced. Then, participants were randomly assigned to study either with the common ground version or the control version of the PCA. After the three loops were completed on a separate interface, participants were redirected to the study. First, they were asked to indicate whether there were any problems when interacting with the chatbot. Then they answered two questions to gauge attention during the chatbot activity.

Next, participants answered questions on their previous usage of chatbots (3 items) and their personal innovativeness (4 items, translated from Van Raaij and Schepers 2008; Cronbach's α = .81). Then, they evaluated the cognitive load (3 items, α = .62; therefore, these items were analyzed separately), the learning materials (2 items) and their judged learning success (1 item), their learning experience with the PCA (3 items, α = .68; items were analyzed separately) followed by questions about how much the chatbot aided their learning (3 items, modelled after McKnight, Choudhury, and Kacmar 2002; α = .80) and an item assessing participants' overall evaluation of the chatbot. Participants in the first data collection wave answered some additional questions. To save time, these items were removed for later data collection waves and will not be considered any further.

To assess experienced common ground, participants were then asked to evaluate the degree of common ground between themselves and the PCA, evaluating how similar their and the PCA's understanding was concerning the study materials (α = .85), the task (α = .77), communication (α = .84), and each other's mental capacities (α = .78). The common ground measures were modelled after Van Rensburg et al. (2022) and consisted of three items for each dimension.

As the main dependent variable, participants were then asked to answer ten questions about the study materials, nine of these were multiple choice questions (with a total of 42 dichotomous items) and one was a

short answer question. Finally, participants answered demographic questions, were given the chance to comment on the study and were thanked for their participation (for an Overview, see Figure 1).

4. Results

To test whether the two versions of the PCA did differ in establishing common ground, we compared judged common ground that participants experienced with the PCA concerning four aspects.[4] For the studied learning materials, participants in the common ground condition (M = 5.20, SD = 0.98) experienced higher common ground with the PCA than participants in the control condition (M = 4.87, SD = 1.21), $t(119) = 1.72$, $p = .044$, $d = 0.30$, 95% CI = [−0.04, 0.65]. However, this was not the case for the other three aspects. Neither participants' judgement on common ground concerning the task, $t(124) = 1.49$, $p = .070$, $d = 0.26$, 95% CI = [−0.08, 0.61], nor their mutual mental models, $t(126) = 0.41$, $p = .342$, $d = 0.07$, 95% CI = [−0.27, 0.42],

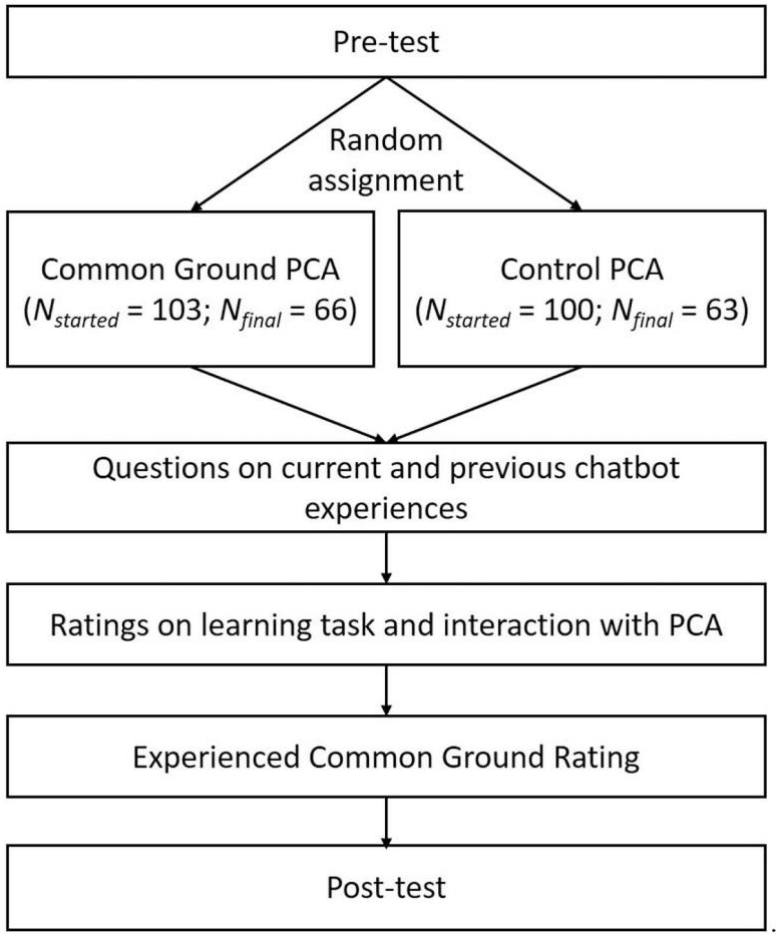

Figure 1. Overview of the procedure.

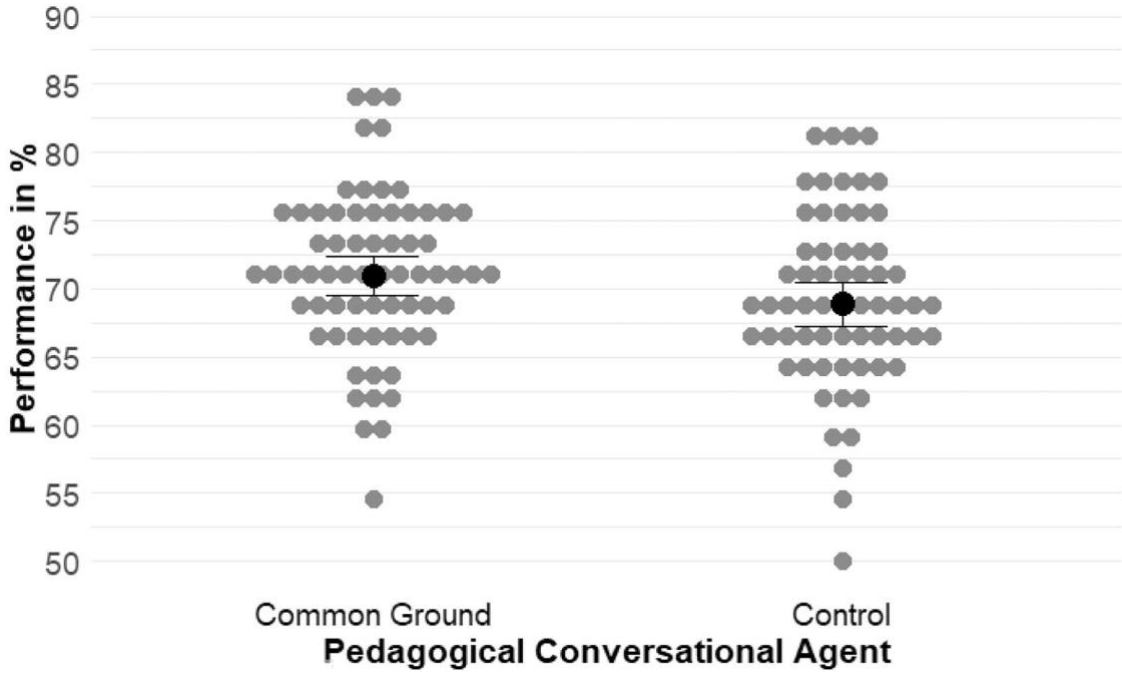

Figure 2. Task performance depending on type of pedagogical conversational agent. Note: Grey dots depict individual test performance proportions. The black dots with error bars indicate group means and confidence intervals around the mean.

nor their common ground on how to communicate, $t(127) = 0.54$, $p = .705$, $d = -0.09$, 95% CI = [−0.44, 0.25], differed significantly depending on the PCA version. Thus, the present manipulation successfully influenced experienced common ground concerning the content of the study materials, which is in line with H1. However, this difference did not generalise to other aspects of common ground beyond the immediate study materials; this might be caused the low conversational flexibility of the present PCAs (see discussion in Section 5.2).

To test the main hypothesis, namely whether grounding improves learning with PCAs, we compared participants post-study test performance depending on PCA version. Participants' test performance was superior for participants in the common ground (M = 70.91%, SD = 5.93) compared to the control PCA (M = 68.87%, SD = 6.50), $t(125) = 1.87$, $p = .032$, $d = 0.33$, 95% CI = [−0.02, 0.68], see Figure 2. Repeating the analysis without excluding participants who spent less than three minutes interacting with the PCA yields similar results, that is test performance was superior for participants in the common ground (M = 70.00%, SD = 6.82) compared to the control PCA (M = 68.33%, SD = 6.84), $t(143) = 1.81$, $p = .037$, $d = 0.30$, 95% CI = [−0.03, 0.63]. To ensure that post-study performance differences were not driven by pre-study knowledge differences, we performed the main analysis on knowledge gains (calculated as post-study minus pre-study test performance). Results were again similar, $t(126) = 2.12$, $p = .018$, $d = 0.37$, 95% CI = [0.03, 0.72]

When including both PCA type (common ground vs. control) and time on task (interaction duration with the PCA) as predictors, PCA type still significantly predicted performance $F(1, 118) = 6.10$, $p = .015$, $h^2_p = .049$, 90% CI = [.005, .126], whereas time on task did not, $F(1, 118) = 0.13$, $p = .720$, $h^2_p = .001$, 90% CI = [.000, .030]. This indicates that the performance improvement caused by the common ground compared to the control PCA is not caused by increased interaction time with the PCA or preexisting knowledge differences. In sum, common ground emphasis by the PCA improved learning success, supporting H2.

Concerning participants' evaluations of the learning task and the PCA, ratings were similar, including participants' overall satisfaction with the PCA, see Table 1.

Finally, interaction durations with the PCA were compared. Participants interacted longer with common ground version (M = 19.56 min, SD = 6.93 min) compared to the control version of the PCA (M = 15.39 min, SD = 5.20), $t(111) = 3.75$, $p < .001$, $d = 0.68$, 95% CI = [0.31, 1.05], see Figure 3.

5. Discussion

In tertiary education, a crucial task for students is determining when their understanding of new information is sufficient, that is, when common ground has been established (Clark and Brennan 1991). The present field experiment tested whether studying with a pedagogical conversational agent that does (vs. does not) explicitly encourage common ground monitoring improves learning. Specifically, in the common ground version, students were repeatedly asked to monitor grounding, deciding whether or not there was a common understanding on key aspects of the study materials. If not, they could initiate repairs and articulate open questions. In the control version, students also decided whether they wanted the additional materials;. only the question wordings differed. We found that both versions of the PCA received similar positive evaluations. Experienced common ground concerning the study materials as well as interaction duration were longer for participants interacting with the common ground (vs. control) PCA. Most importantly, in a post-learning test, performance was superior for participants who had studied with the common ground compared to the control version of the PCA. This learning advantage was not due to time on task, as the PCA version predicted learning success even when interaction duration was controlled, so that the learning advantage caused by the common ground emphasis is not merely caused by students' longer study times. In sum, although both versions of the PCA provided the same texts, quiz questions, and, upon request, the same additional information, interacting with the PCA that encouraged (vs. did not actively encourage) grounding improved learning.

5.1. Common ground in teaching

Dyadic conversations usually run very smoothly (Garrod and Pickering 2004). One reason for this is continual grounding; both partners monitor common ground, provide backchannels and frequently initiate repairs. In a large linguistically-diverse sample of conversations, for example, repairs have been found to occur on average every 1.4 minutes (Dingemanse et al. 2015). Thus, in typical conversations, grounding is a frequent and collaborative occurrence. This is also the case in dyadic teaching situations, where, for example, teachers sometimes ask students to introduce new reference

terms (Turner and Knutsen 2021). However, in university teaching, lectures are the norm and the majority of students never initiate a conversational repair, so that grounding mainly consists of audience design performed by the instructor. Studying with PCAs, in contrast, can be much more interactive,

Table 1. Evaluations of the learning experience and the pedagogical conversational agent.

Construct	Mean (SD) common ground PCA	Mean (SD) control PCA	*t*-test	*p*-value unadjusted, *p*-value Bonferroni-adjusted	Cohen's *d*, 95% CI
Content complexity	3.53 (0.83)	3.37 (1.00)	$t(120) = 1.02$	$p = .31$, $p = 1.00$	$d = 0.18$, [−0.17, 0.53]
Judgment learning success	3.30 (0.89)	3.33 (0.74)	$t(125) = 0.21$	$p = .83$, $p = 1.00$	$d = -0.04$, [−0.38, 0.31]
Cognitive load: required	3.64 (0.80)	3.38 (0.77)	$t(127) = 1.85$	$p = .07$, $p = 0.73$	$d = 0.33$, [−0.02, 0.67]
Cognitive load: invested	3.33 (0.88)	3.14 (0.72)	$t(124) = 1.35$	$p = .18$, $p = 1.00$	$d = 0.24$, [0.11, 0.58]
Cognitive load: difficult	3.14 (0.84)	3.02 (0.75)	$t(126) = 0.86$	$p = .39$, $p = 1.00$	$d = 0.15$, [−0.19, 0.50]
Judgment of learning (in %)	69.12 (16.71)	69.25 (17.15)	$t(126) = 0.04$	$p = .97$, $p = 1.00$	$d = 0.00$, [−0.35, 0.34]
Learning with chatbot: pleasant	3.53 (0.95)	3.75 (0.92)	$t(127) = 1.31$	$p = .19$, $p = 1.00$	$d = -0.23$, [−0.58, 0.12]
Learning with chatbot: motivating	3.17 (1.10)	3.29 (0.96)	$t(126) = 0.66$	$p = .51$, $p = 1.00$	$d = -0.12$, [−0.46, 0.23]
Learning with chatbot: difficult	2.56 (0.98)	2.60 (0.96)	$t(127) = 0.25$	$p = .80$, $p = 1.00$	$d = -0.04$, [−0.39, 0.30]
Chabot as learning aid	3.63 (0.77)	3.50 (0.87)	$t(124) = 0.89$	$p = .38$, $p = 1.00$	$d = 0.16$, [−0.19, 0.50]
Overall chatbot evaluation	3.71 (0.70)	3.73 (0.87)	$t(119) = 0.13$	$p = .90$, $p = 1.00$	$d = -0.02$, [−0.37, 0.32]

Note: Judgment of Learning was judged in percent (0%–100%). All remaining judgments were performed using 1–5 rating scales with higher values indicating higher judged prevalence or higher satisfaction.

with grounding depending on both instructors and learners, resembling rather dyadic conversations than lectures. Thus, employing PCAs to replace lectures can shift grounding from almost exclusively resting on audience design by the instructor to a more balanced interaction, with greater weight on backchannels and repairs, initiated by the student.

However, the present research indicates that, unless prompted to do so, students do not make full use of the interactive possibilities and do not fully monitor grounding. Common ground concerning the learning materials was experienced to be lower and learning was less successful for students studying with the control PCA, which did not explicitly ask students to monitor and repair common ground. Thus, we find that even in university education, monitoring common ground when interacting with a digital tutor does not fully occur without additional prompting. Therefore, a conclusion of the present research for designing pedagogical conversational agents is that they should incorporate grounding measures. Students should be actively encouraged to monitor and, where necessary, repair common ground.

5.2. Limitations and future research

A question that the present research does not address is generalizability. We only examined immediate test performance (ca. 5 min after the learning activity ceased) and therefore cannot tell whether the learning advantage of the common ground manipulation is constant over time and also holds for delayed testing (Soderstrom and Bjork 2015). Similarly, our final test did not distinguish between knowledge and transfer questions. Most questions contained some transfer aspect; however, the degree varied. To gain a more thorough understanding of the benefits of grounding by PCAs, different types of test questions and different time delays should be compared in future research (Schweppe and Rummer 2016; Schweppe, Eitel, and Rummer 2015).

Additionally, to intensify common ground establishment, future conversational agents should use more advanced language models (Adıgüzel, Kaya, and Cansu 2023; Giannakos et al. 2025; Yan et al. 2024). In the present research, we used a PCA with intent recognition, which responded to user queries by following predefined rules or conversational flows (Schöbel et al. 2024). From an internal validity point of view, a predefined conversational flow enables a confound-free manipulation. This enables the present demonstration that common ground facilitation by merely asking participant to reflect on grounding and, if necessary, initiate repairs, improves grounding and learning success. Therefore, we can conclude that the present grounding manipulation facilitated performance differences.

However, this gain in internal validity has also disadvantages. For teaching situations, having to pre-specify all reasonable conversational paths is cumbersome and restricts conversations. This might also be the reason why participants' evaluations about

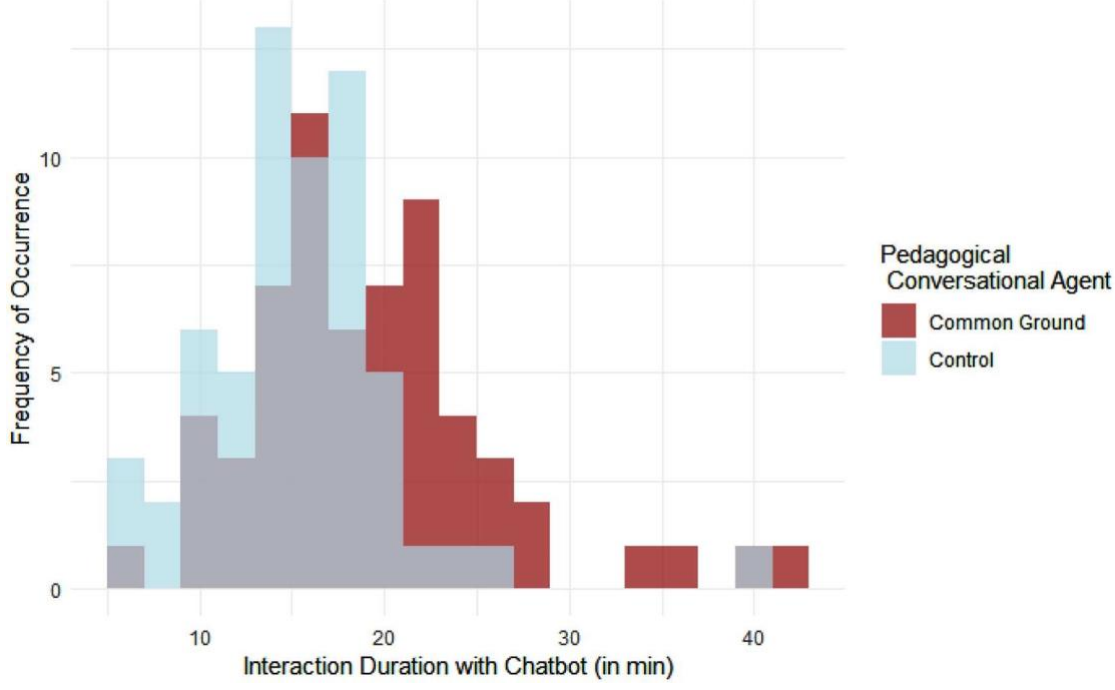

Figure 3. Histogram depicting individual interaction duration depending on type of pedagogical conversational agent. Note: Light (blue) areas indicate number of participants in the Control PCA condition, dark (red) areas indicate number of participants in the Common Ground PCA condition; medium coloured areas indicates number of participants of both PCAs.

the extent of knowing each other's mental capacities as well as common ground concerning conversation strategies and learning tasks did not differ between the two PCAs. The conversations were impoverished so that probably no representations of mental capacities or shared understanding about the task or conversations could develop with either PCA.

The integration of large language models would enable more human-like conversations (Díaz and Nussbaum 2024), which would enable a more interactive conversational style and more flexible distribution of grounding (for measures, see Bovet, Knutsen, and Fossard 2024). We hypothesise that this will lead to larger effects of common ground facilitation, as additional grounding aspects could be incorporated, such as repairs performed by students. As self-generated text has been found to be better recalled that other-generated text (Jacob, Lachner, and Scheiter 2020; Wang et al. 2023), we assume that repairs that are generated (vs. merely received) by students will facilitate learning (for related findings, see Müller, Paul, and Li 2021), strengthening the influence of grounding interventions. Thus, the usage of large language models would enable common ground interventions that incorporate multiple aspects of common ground. If these are successful, we would expect a common ground PCA to yield higher subjective common ground judgments on various dimensions, including shared mental representations and common ground concerning preferred conversation strategies and tasks. Whether this is indeed the case needs to be examined by future research, which could also dig deeper into this matter by employing qualitative methods to explore students' experiences during common ground formation.

Another open question concerns the generalizability of the present findings beyond the present topic (see Dunlosky et al. 2013). Here, we employed one specific topic in an introductory psychology class. As the grounding manipulation is content-independent, it can be easily applied to most topics, and we would predict similar effects for most complex topics in higher education. Previous research demonstrated learning advantages after studying with PCAs (compared to a control activity) across various subjects (Kuhail et al. 2023), so that we would also predict similar effects across various subjects.

However, generalizability to other student populations is less clear to predict. PCAs might be especially effective in tertiary education (Wu and Yu 2024; for the opposite result, see Schroeder, Adesope, and Gilbert 2013), so that the effectiveness of at least some communicative behaviour seems to depend on educational level. Accordingly, the present grounding manipulation might only be helpful for conversational agents whose target group are advanced learners. In addition to educational level, other student characteristics, such as special needs, might also predict how much students profit from a grounding intervention. PCAs can be adapted to help students with special needs, for example, by training specific skills (Torrado, Bakke, and Gabarron 2023). Students with autism spectrum disorder might profit more strongly than other students from common ground elements in PCAs[5] because students with autism spectrum disorder have been found to take longer to establish common ground (Karin et al. 2025) so that additional grounding elements might be especially efficacious.

Future research needs to confirm the generalizability of the present findings beyond the present study conditions, using a wider variety of topics, recall intervals, learning questions, and student populations. However, from the present research, we predict that incorporating questions that foster grounding performed by students will benefit students in higher education by facilitating increased learning. The present field experiment indicates that in real-world study situations, grounding interventions can facilitate student engagement and learning.

In addition to study situations, we hypothesise a beneficial effect of grounding will also occur in other types of interaction with asymmetric knowledge where accuracy is important. As conversational agents are becoming increasingly integrated into everyday life (Schöbel et al. 2024), collaboration between humans and conversational agents is becoming ever more frequent, highlighting the need for a thorough understanding of behaviours that facilitate communication (Seeber et al. 2020a; 2020b). Repairs have been found to be seldom when miscommunication does not deteriorate task performance (Galantucci et al. 2020). Similarly, affiliation- oriented compared to task-oriented conversions have been found to elicit fewer repairs (Dideriksen et al. 2023). Thus, in human conversation, repairs are less frequent when they are deemed nonessential for the purpose of conversation. However, when important for the task and when the two parties of a conversation differ in knowledge, repairs are important for the conversation to be efficient and successful (Isaacs and Clark 1987). Under these conditions, grounding is beneficial to task performance so that a manipulation that asks users to monitor and

repair common ground should improve task performance – not only for human – human but also for human – computer interactions. The present research lends first evidence in the domain of higher education.

6. Conclusion

A long-term aim for the development of pedagogical conversational agents is creating digital learning tutors that facilitate learning as much as human tutors (VanLehn 2011). If this goal is reached, PCAs could be widely implemented, enriching both traditional and online teaching. Ideally, this will further student retention and learning success without requiring large additional resources, making good teaching affordable and consequently more widely accessible (Holmes, Bialik, and Fadel 2020; Song, Oh, and Rice 2017; Yan et al. 2024). To reach this goal, it is vital to determine how digital tutors should behave – especially how they should communicate – in order to improve learning (Ramachandran, Sebo, and Scassellati 2019). The present research contributes to this goal by examining common ground, and more specifically, the influence of the PCA's strategically encouraging users to monitor grounding by asking for acknowledgments or repair initiations. We observed that students studied more successfully with the common ground compared to an otherwise identical control version of a PCA. Thereby we show that, and how, grounding principles should be incorporated in pedagogical conversational agents.

Notes

1. The high non-completion rate might have been caused by the study context. The easiest way to learn all course materials consisted in interacting with the PCA. Participants who did not wish to volunteer for the study were informed that they could simply discontinue at any time after the learning activity.
2. Of these, eight restarted the study after the learning task; these were excluded from analyses concerning time on task but not from any other analyses.
3. For a description of the design principles employed, see Tolzin et al. (2023).
4. As our hypotheses were directional, i.e., increases in experienced common ground and improved learning for the common ground compared to the control PCA, the hypotheses tested one-tailed. All other tests were two-tailed.
5. We thank an anonymous reviewer for this idea.

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Author contributions

CRediT: **Anita Körner:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft; **Antonia Tolzin:** Conceptualization, Methodology, Software, Visualization, Writing – review & editing; **Andreas Janson:** Conceptualization, Methodology, Writing – review & editing; **Jan Marco Leimeister:** Funding acquisition, Writing – review & editing; **Ralf Rummer:** Conceptualization, Funding acquisition, Writing – review & editing.

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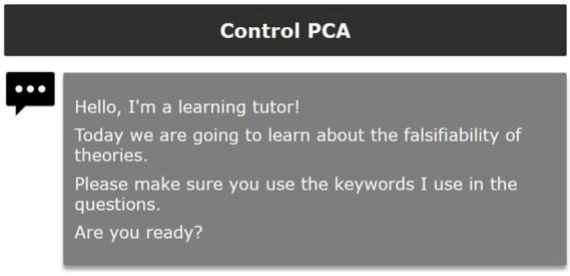
Appendix



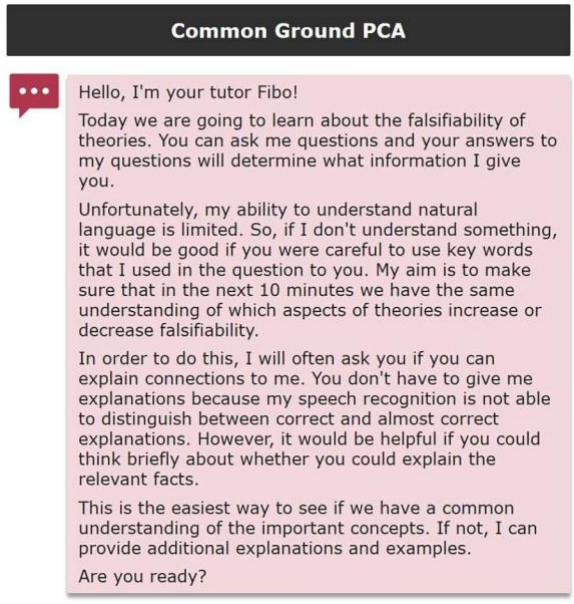


Figure A1. Introduction Depending on Type of Pedagogical Conversational Agent. Note: All texts are translations of the originals.

Figure A2. Questions on Whether or Not the Pedagogical Conversational Agent Should Provide Additional Explanation Depending on PCA Type. Note: All texts are translations of the originals. The common ground PCA asks for backchannels and, if necessary, common ground repair initiations.

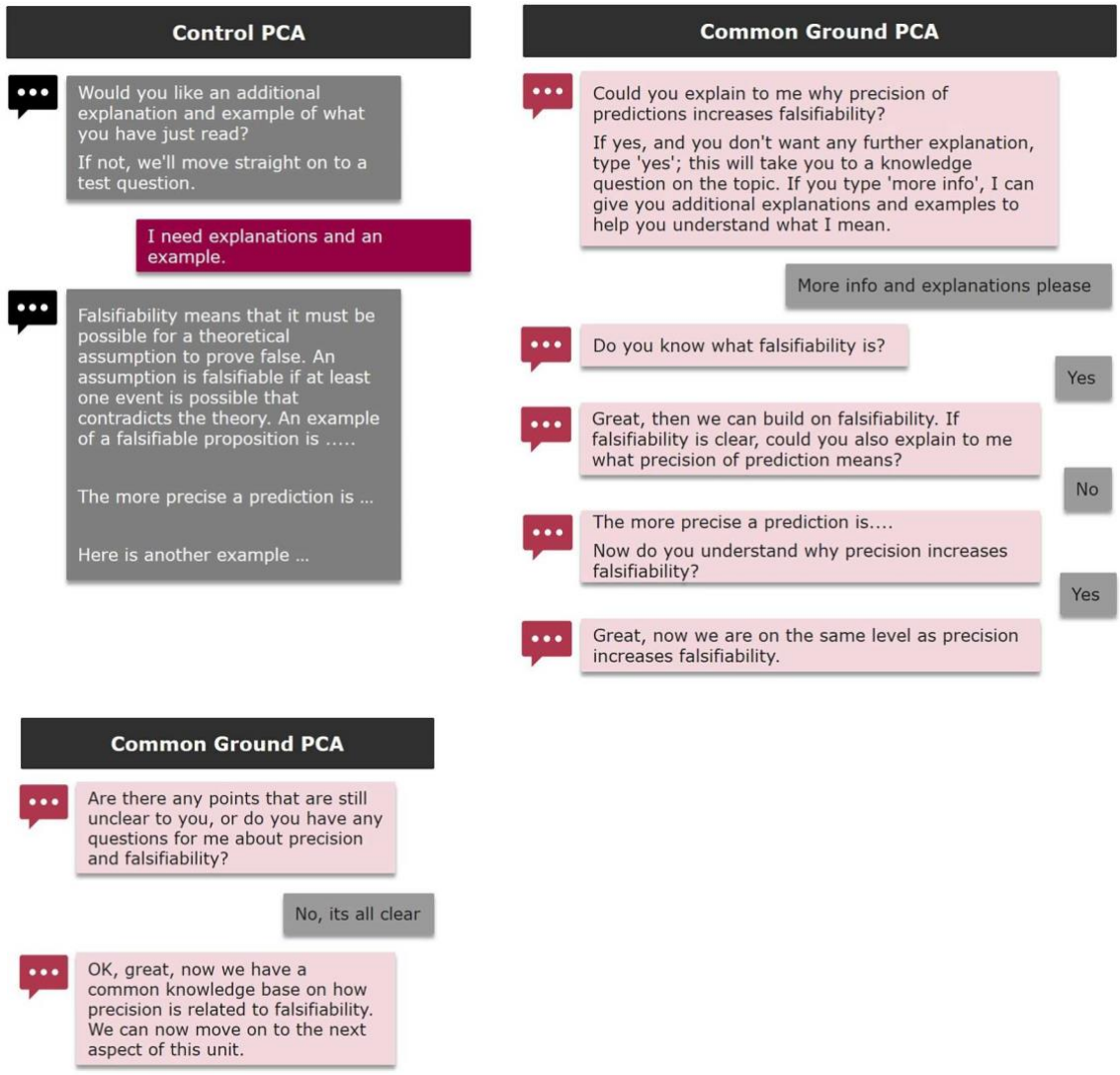


Figure A3. Final Question in Each Loop: Participants in the Common Ground Conditions Could State Open Questions. Note: All texts are translations of the originals. The common ground PCA asks participants to monitor and articulate gaps in the common ground.

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